Sound speed and resonance

antocciphysics | TEACHER KEY | CPA

Learning target: Use travel time and resonant conditions to interpret sound.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 6; 3 course days later.*
The 5 cm wave generally spreads more because wavelength is larger relative to the same opening.

02. Explain a worked example (worked)
Half a wavelength fits; displacement node near the middle and antinodes at open ends.

03. Look and explain (model)
Sound travels out and back; total 2×range.

04. Build the solution (practice)
40 m total; 20 m range.

05. Use the idea (reason)
Natural modes set preferred frequencies; response also depends coupling and damping.

06. Choose a method (mixed) *Review: Unit 5, Day 9; 12 course days later.*
30% efficiency; 420 J rejected, assuming a complete cycle.

07. Exit check (exit)
30 m.

08. STRETCH: Apply it another way (stretch)
340 and 170 Hz; doubling length halves fundamental frequency.

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01. Bring back an earlier idea (retrieve) *Review: Unit 6, Day 6; 3 course days later.*
Wavelength becomes 1/3; typically less spreading when the opening is large relative to the new wavelength.

02. Explain a worked example (worked)
Half a wavelength fits; displacement node near the middle and antinodes at open ends.

03. Represent and interpret (model)
Sound travels out and back; total 2×range.

04. Solve and justify (practice)
51 m;λ1 m; f 340 Hz.

05. Analyze the reasoning (reason)
Natural modes set preferred frequencies; response also depends coupling and damping.

06. Choose a method (mixed) *Review: Unit 5, Day 9; 12 course days later.*
Work 1050 J; efficiency 30%.

07. Exit check (exit)
340 Hz.

08. STRETCH: Apply it another way (stretch)
Fundamental 100 Hz; odd harmonics 100, 300, 500 Hz.